

CO to PO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2											2	
CO2	3	2	3	3									2	
CO3	3	2	3	3									2	
CO4	3	2	3	3									2	
CO5	3	3	3	3					3	3		3	3	
Total	15	11	12	12					3	3		3	11	
Score	3	3	3	3					3	3		3	3	

COURSE CODE	COURSE TITLE	L	T	P	E	C
UCS2503	SOFTWARE ENGINEERING	3	0	0	0	3

OBJECTIVES

- To understand the phases in a software project, estimate cost and effort.
- To understand fundamental concepts of requirements engineering and Analysis Modeling.
- To understand the various software design methodologies
- To learn various testing techniques and maintenance measures.
- To understand agile development and Devops

UNIT I SOFTWARE PROCESS AND PLANNING

9

Introduction to Software Engineering; Objectives, Principles and Practices; The Software Development Life Cycle : Pre-development phases of the SDLC – Development specific phases of the SDLC – Post-development phases of the SDLC; Methodologies Paradigm and Practices : Process methodologies – Development paradigms – Development practices; Project Planning Process; Software Project Estimation: Decomposition techniques – Empirical estimation models – The make/buy decision – Project scheduling; Risk Management; Handling Ethical Dilemmas.

UNIT II REQUIREMENTS ANALYSIS AND SPECIFICATION

9

Software Requirements: Functional and non-functional – Security requirements – User requirements – System requirements – Software requirements document; Requirement Engineering Process: Feasibility studies – Requirements elicitation and analysis – Requirements validation – Requirements Management; Classical Analysis: Structured system analysis; Requirement modelling tools.

UNIT III SOFTWARE DESIGN

9

Design Concepts: Design process – Design concepts – Modularity, Coupling and cohesion – Design model – Modeling principles; Structured Design; Architectural Design: Architectural styles; Architecture for Network based Applications – Decentralized Architectures.

UNIT IV SOFTWARE TESTING

9

Software Testing Fundamentals; Internal and External Views of Testing: White box testing – Basis path testing – Control structure testing– Black box testing – Unit testing – Integration testing – Regression testing – Validation testing – System testing – Security testing; Testing Tool; Debugging; Software Implementation: Coding Practices and Principles; Maintenance: Types.

UNIT V AGILE DEVELOPMENT AND DEVOPS

9

Agile Development: Agile Teams – Team and Scrum – Branches – Pull Requests – Reviews – Integration – Agile Iterations – Reporting and fixing bugs; Dev/Ops: From development to deployment – Three-Tier – Responsiveness, Service level objectives, and Apdex – Releases and feature flags – Monitoring and finding bottlenecks – Improving rendering and database performance with caching; Security: Defending customer data in application.

TOTAL PERIODS: 45

COURSE OUTCOMES

On successful completion of this course, the student will be able to:

1. Choose an appropriate process model and estimate project cost and effort required by applying software engineering principles(K5)
2. Identify and analyze the requirements and construct their models (K5)
3. Apply systematic procedure for software design (K3)
4. Compare and contrast the various testing and maintenance activities (K2)
5. Make use of agile development and Devops (K3)
6. Identify unethical issues and apply ethical practices for a given case study (K3)

TEXTBOOKS

1. Roger S Pressman, Bruce R Maxin “Software Engineering – A Practitioner’s Approach”, McGraw Hill International Edition, Eighth Edition, 2015.
2. Armando Fox and David Patterson, Engineering Software as a Service: An Agile Approach Using Cloud Computing”, Strawberry Canyon LLC, Second Beta Edition, 2021.

REFERENCE BOOKS

1. Ian Sommerville, “Software Engineering”, Pearson Education Asia,Tenth Edition, 2015.
2. Stephen R Schach, “Software Engineering”, Tata McGraw-Hill Publishing Company Limited,2007.
3. Brian Albee, Hands-On Software Engineering with Python, Packt Publishing, 2018.
4. Kelkar S A,“Software Engineering”, Prentice Hall of India, 2007.

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CO3	3	2	2										2	2
CO4	3	2	2										3	3
CO5	3													
CO6	3							3		3		1		
Score	18	9	7		3	2	3	3	3	6	3	1	7	11
Course Mapping	3	3	3		3	2	3	3	3	3	3	1	3	3